DATA ANALYTICS BY DURGA

Data Analytics Assignments

Excel Dashboards • Kaggle Datasets • Power Query Merge Types

General Instructions

- Complete three dashboard assignments in Excel using the Kaggle datasets listed below.
- Keep an untouched copy of the downloaded source data. Document cleaning, filtering, and transformations.
- Use Excel Tables, PivotTables/PivotCharts, and slicers where supported. Each dashboard should include 4–6 KPIs, at least 3 charts, and at least 2 useful slicers.
- Create an Insights sheet with five evidence-based findings and two recommendations. Avoid claims the data cannot support.
- Submit the Excel workbook, a PDF or screenshots of the dashboard, and a short note explaining the data preparation and findings.
- Record the dataset title, Kaggle URL, download date, and any limitations.

Assignment 1: Sports Analytics Dashboard

Objective: Analyze player attributes and compare patterns across age, nationality, club, position, and performance.

Suggested dataset: FIFA 22 Complete Player Dataset — https://www.kaggle.com/datasets/stefanoleone992/fifa-22-complete-player-dataset

Required analysis

- Player count, average age, average overall rating, and average potential rating where available.
- Top 10 players by overall rating or another clearly defined performance metric.
- Player count by nationality, club, and playing position.
- Player distribution by age group or rating band.
- Compare age with overall rating and discuss limitations of interpreting potential as actual performance.

Suggested visuals and slicers

- Horizontal bar chart: Top 10 players, clubs, or nationalities.
- Column chart: player count by position or age group.
- Scatter plot: age vs. overall rating if practical.
- Slicers: nationality, club, position, preferred foot, or league where available.

Assignment 2: Sales Analytics Dashboard

Objective: Analyze sales performance, product/category comparisons, customer segments, and transaction patterns.

Suggested dataset: Supermarket Sales — https://www.kaggle.com/datasets/aungpyaeap/supermarket-sales

Required analysis

- Total sales/revenue, transaction count, average transaction value, and quantity sold where available.
- Sales by product line/category, branch/store, city/region, and payment method.

- Sales trend by date, month, or day of week where supported.
- Compare customer types or segments and identify top/bottom categories using a stated metric.

Suggested visuals and slicers

- Line chart: sales over time.
- Bar chart: sales by product line/category or branch.
- Column chart: transaction count or sales by payment method/customer type.
- Slicers: date/month, branch, product line/category, customer type, or payment method.

Caution: Do not call gross income net profit unless the dataset supports that interpretation. Confirm field definitions.

Assignment 3: Financial Analytics Dashboard

Objective: Summarize transaction patterns and, where labels exist, compare fraudulent and non-fraudulent activity.

Suggested dataset: Credit Card Fraud Detection — https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Required analysis

- Transaction count, total transaction amount, average/median transaction amount where available.
- Fraud count and fraud percentage if a fraud label is included.
- Compare transaction amount distributions for labelled classes using appropriate aggregations.
- Trend by time period if timestamp fields are available.
- Explain anonymized/coded columns; do not assign meanings to variables without documentation.

Suggested visuals and slicers

- KPI cards: transaction count, total amount, flagged count, and flagged percentage where applicable.
- Bar chart: counts or amounts by category if available.
- Line chart: transaction volume over time if timestamps are available.
- Slicers: time period, transaction category, region, or fraud label if supported.

Use only the anonymized public dataset. Avoid publishing row-level data or sensitive information unnecessarily.

Common Dashboard Requirements

Area	Requirement
Data preparation	Check missing values, duplicates, data types, and inconsistent formats; document decisions.
KPIs	Include 4–6 relevant metrics with clear labels and appropriate number formats.
Visuals	At least 3 charts with descriptive titles and readable axes.
Interactivity	At least 2 slicers where supported; connect them to relevant PivotTables/PivotCharts.
Design	Consistent colors, aligned objects, readable fonts, and no unnecessary 3-D effects.
Insights	Five findings and two practical recommendations grounded in the data.
Refreshability	Use Power Query/PivotTable refresh where appropriate and note limitations.

Assignment 4: Power Query — Merge Types Using Olist

Objective: Import multiple related CSV files from the Brazilian E-Commerce dataset and demonstrate different Merge Queries join types in Excel Power Query.

Dataset: Brazilian E-Commerce Public Dataset by Olist — https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Download the dataset ZIP from Kaggle and extract the CSV files before importing. The dataset includes related tables for orders, customers, order items, payments, reviews, products, sellers, geolocation, and category translation.

Suggested tables and matching keys

Tables	Key	Possible analysis
Orders + Customers	customer_id	Customer location and order information
Orders + Order Items	order_id	Order-level and item-level analysis
Orders + Payments	order_id	Payment methods and payment values
Orders + Reviews	order_id	Review score and order status
Order Items + Products	product_id	Product categories and item prices
Order Items + Sellers	seller_id	Seller location and item sales
Products + Category Translation	product_category_name	Readable product category labels

Required Power Query tasks

1. Import at least three Olist CSV files as separate queries using Data → Get Data → From File → From Text/CSV.
2. Check column names and data types. Make matching key columns compatible before merging.
3. Perform a Left Outer Join on Orders and Customers using customer_id; expand selected customer fields.
4. Perform an Inner Join on Orders and Order Items using order_id. Record the output row count and explain why orders may repeat when they contain multiple items.
5. Perform a Full Outer Join on two related tables and identify unmatched rows using nulls in expanded columns.
6. Perform a Left Anti Join to find rows in the first table without a match in the second; clearly state which table is first.
7. Perform a Right Anti Join to find rows in the second table without a match in the first.
8. Perform a Right Outer Join and compare it with a Left Outer Join after reversing the table order, where appropriate.
9. Load at least two results to separate Excel worksheets and name the sheets clearly.
10. Create a comparison table describing the purpose and output of each join type.

Power Query Join Reference

Join kind	Expected result
Left Outer	All rows from the first/top table and matching rows from the second/bottom table.
Right Outer	All rows from the second/bottom table and matching rows from the first/top table.
Inner	Only matching rows from both tables.
Full Outer	All rows from both tables, matched where possible.
Left Anti	Rows from the first table with no match in the second.
Right Anti	Rows from the second table with no match in the first.

Important Olist Data Considerations

- Orders and order items are not one-to-one. One order can contain multiple items, so a merge can increase row count.
- Payments can contain multiple records per order. Do not assume every table has one row per order.
- Use order_id, customer_id, product_id, and seller_id only for the relationships they represent.
- Inspect null keys and unmatched records. Do not assume blank keys identify the same entity.
- Joining geolocation on a partial postal-code prefix can match multiple rows; explain the implications.
- Do not sum order-level amounts after merging to item-level rows without checking for duplication and choosing the correct level of analysis.
- Follow the attribution and license terms shown on the Kaggle dataset page.

Submission Checklist

- Three Excel dashboards or clearly separated dashboard sheets: sports, sales, and finance.

- A Power Query workbook or named queries demonstrating Olist merge exercises.
- An Insights worksheet for each dashboard.
- PDF exports or screenshots of the completed dashboards.
- A short reflection describing cleaning, merges, findings, and limitations.

Evaluation Rubric (100 Marks)

Evaluation area	Marks
Data import and cleaning	15
Correct PivotTables, summaries, and KPIs	15
Chart selection and readability	15
Dashboard layout and visual design	10
Slicers and interactivity	10
Insights and recommendations	10
Power Query merge types and key selection	15
Validation, documentation, and submission quality	10

Academic integrity: Students may discuss concepts, but each submission must reflect their own data preparation, analysis, design choices, and interpretation. Cite the dataset source and disclose templates or external assistance.